

Customization of the CVA6 Core

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Abstract

The CVA6 is a low-power, open source CPU written in SystemVerilog. In the team, we use it to run a small hypervisor, Fork-Nox. The goal of this project is to identify, quantify and if possible implement microarchitectural characteristics that would decrease the overhead of Fork-nox compared to native execution. One lead is to study the influence of data / instruction cache sizes, branch predictor, frequency and TLB size. Another lead is to modify the granularity of virtual-to-hypervisor interception mechanisms, which requires deeper modification of the code base.

1. Usage Summary

The first part of this project consisted in evaluating the resource usage due to different parameters of the CVA6 architecture. For evaluation purposes a baseline configuration is given with the following parameters.

Table 1. Baseline Bitstream Configuration

Parameter	Size
Data Cache	4096[bytes]
Data Cache Associativity	4
Data Cache Line Width	128
Instruction Cache	4096[bytes]
Instruction Cache Associativity	4
Instruction Cache Line Width	128
BTB Entries	16
BHT Entries	16

The CVA6 architecture was then implemented in a Pynq-Z2 FPGA. The resources of the FPGA can be mainly separated into 4 groups which are:

- Flip Flops (**FF**)
- Look up Tables (**LUT**)

- Digital Signal Processor (**DSP**)
- Block RAM (**BRAM**)

To start let's see the resource usage of the baseline bitstream.

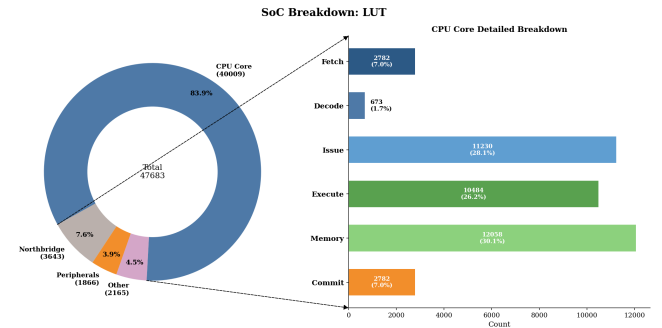


Figure 1. Baseline LUT Usage

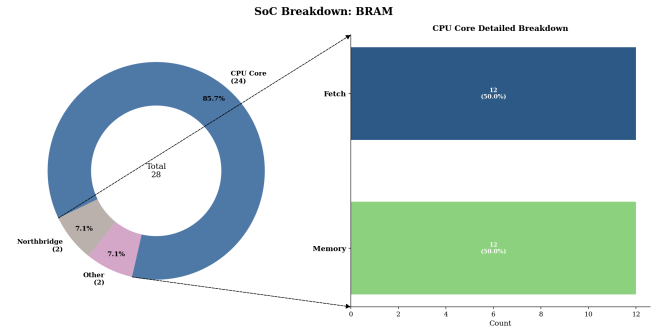


Figure 2. Baseline BRAM Usage

For visualization purposes the following fields have been included in the *other* category:

- SPI Controller: The interface used for the SD Card or Flash memory.
- UART: The serial interface used for the console/terminal.
- CLINT (Core Local Interruptor): The timer module responsible for generating software and timer interrupts.
- BootROM: The small Read-Only Memory containing the very first instructions the CPU executes upon reset.

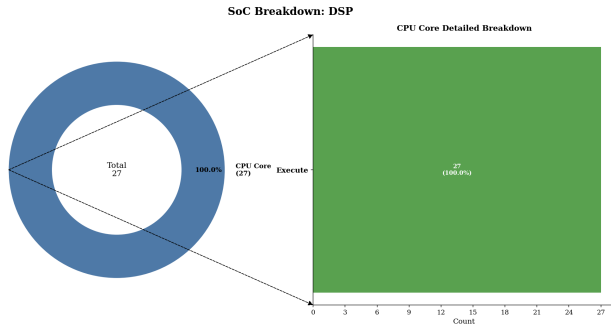


Figure 3. Baseline DSP Usage

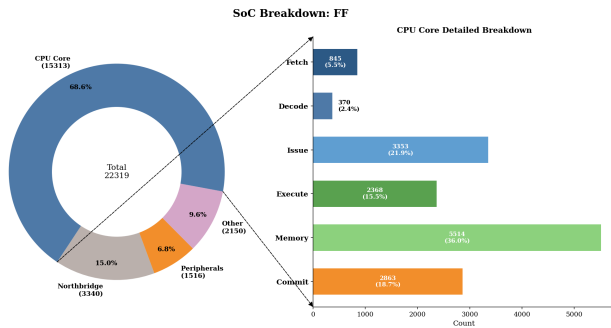


Figure 4. Baseline FF Usage

- Memory Interconnect: The AXI adapters and shims that handle width or clock conversion between the CPU and the main memory bus.
- Glue Logic: The calculated remainder (SoC Total Everything Else). This represents the top-level wires, clock buffers, and reset synchronizers that connect all these modules together.

Also the calculation of the 6 stages involved the following modules from the utilization reports obtained from Vivado.

Table 2. Utilization Report Modules used for Usage Calculation

Stage	Modules
Fetch	i_frontend + i_cva6_icache
Decode	id_stage_i
Issue	issue_stage_i
Execute	ex_stage_i
Memory	lsu_i + i_wt_dcache
Commit	i_scoreboard